differential form of Maxwell's equation is beyond higher secondary level. So we focus only the integral form of Maxwell's equations.

First equation

It is nothing but the Gauss's law of electricity. It relates the net electric flux to net electric charge enclosed in a surface. Mathematically, it is expressed as

$$\oint_s \vec{E}\cdot d\vec{A} = \frac{Q_{enclosed}}{\epsilon_\circ}$$

(Gauss's law for electricity) (5.7)

where $\vec{E}$ is the electric field and Q enclosed is the net charge enclosed by the surface S. This equation is true for both discrete and continuous distribution of charges.

It also indicates that the electric field lines start from positive charge and terminate at negative charge. This implies that the electric field lines do not form a continuous closed path. In other words, it means that an isolated positive charge or negative charge can exist.

Second equation

This law is similar to Gauss's law for electricity. So this law can also be called as Gauss's law for magnetism. The surface integral of magnetic field over a closed surface is zero. Mathematically,

$$\oint_s \vec{B}\cdot d\vec{A} = 0$$

(Gauss's law for magnetism) (5.8)

where $\vec{B}$ is the magnetic field.

This equation implies that the magnetic lines of force form a continuous closed path. In other words, it means that no isolated magnetic monopole exists.

Third equation

It is Faraday's law of electromagnetic induction. This law relates electric field with the changing magnetic flux which is mathematically written as

$$\oint_l \vec{E}\cdot d\vec{l} = -\frac{d}{dt}\Phi_B \quad \text{(Faraday's law)} \qquad (5.9)$$

where $\vec{E}$ is the electric field. This equation implies that the line integral of the electric field around any closed path is equal to the rate of change of magnetic flux through the closed path bounded by the surface.

Our modern technological revolution is due to Faraday's laws of electromagnetic induction.

Fourth equation

It is modified Ampere's circuital law. This is also known as Ampere – Maxwell law. This law relates the magnetic field around any closed path to the conduction current and displacement current through that path.

$$\oint_l \vec{B}.\vec{dl} = \mu_0 i_C + \mu_0\epsilon_0 \frac{d}{dt}\oint_s \vec{E}.\vec{dA}$$

(Ampere-Maxwell law) (5.10)

where $\vec{B}$ is the magnetic field. This equation shows that both conduction current and displacement current produce magnetic field.

These four equations are known as Maxwell's equations in electrodynamics. This equation ensures the existence of electromagnetic waves. The entire communication system in the world depends on electromagnetic waves. In fact our understanding of stars, galaxy, planets etc come by analysing the electromagnetic waves emitted by these astronomical objects.

UNIT-5(XII-Physics_Vol-1).indd 271 14-02-2020 20:11:21